Developer Workstation - Configuring Your Eclipse IDE

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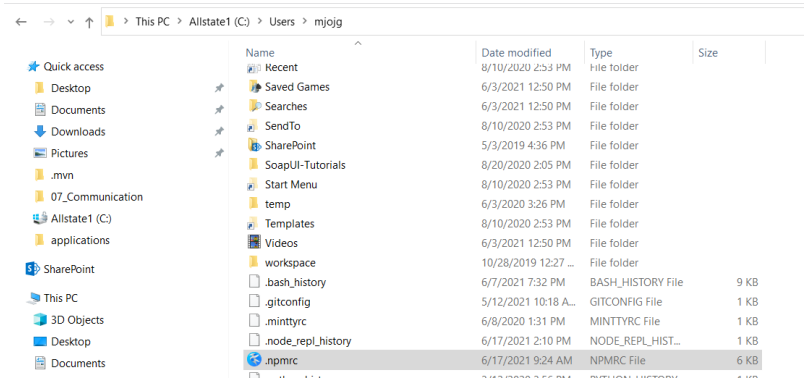
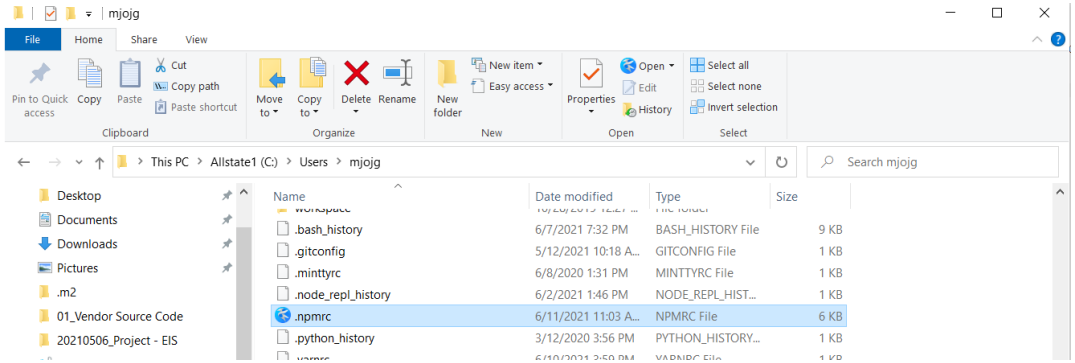
Scope of Page: (Currently in DRAFT Form and should be considered a fluid document)

To provide any given developer the necessary steps to stand up their Developer Workstation for this effort.

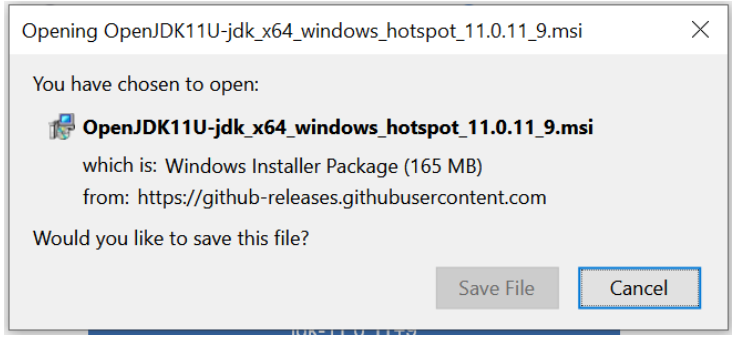
Steps	Instructions	Description / Comments
1	Request Elevated Rights on Developer Laptop	Submit Service NOW Ticket to request ADMIN Rights to your LOCAL MACHINE. Follow the steps below: 1. Click on the following Service NOW Link a. **Description:** Local Desktop Administration - Add User 2. Fill out the following information: a. **Business Unit**: Allstate Technology and Strategic Ventures b. **Division**: Infrastructure Services & Innovation c. **Department**: Infrastructure Applications d. **Location**: South e. **Machine Name**: Either found at “MyComputer” or found by typing “HOSTNAME” at the command prompt f. **Business Need**: New developer resource with Allstate Benefits Effort, <*entertEffortNameHere*>, here in Jacksonville, Florida. Require either FULL Administrative Rights or Elevated R
2	Download **Node.js**	Versions, **14.15.0** can be found at the following link: https://nodejs.org/download/release/v14.5.0/ I chose to download the Windows 64 bit Version as an MSI File. ******It is critical that Step 1 was accomplished first in order to run this MSI File******
3	Right-Click on the **Node.js** MSI File and "**Run Elevated with DefendPoint**"	This will walk through a typical windows install process. The only nuance that I chose to update the install path to "**C:\nodejs**" from the default. "**C:\Program Files\nodejs**". Run this to successful completion.
4	Create a shortcut on your desktop for the "**Node.js**" Command Prompt	Found beneath folder structure: **C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Node.js**
5	Update the "**PATH**" System Environment Variable for Node.js	This is to compensate and validate that the value is pointing at the destination folder set in **Step #3.**
6	Create the following System Environment Variables for Node.js Installs: a. NODE_TLS_REJECT_UNAUTHORIZED = 0 b. GIT_SSL_NO_VERIFY = 1	***This is a pre-requisite.*** In order to perform **Node.js Package Installs using the "npm" a Node Package Manager**, at the **Command Prompt, Node.js** EXPECTS a Signed Certificate from a qualified Certificate Au Unfortunately when first installing packages from the "**Node.js" Command Prompt,** the developer may not have such certificates LOCAL to their machine. These System Environment Variables provide the necessary means around that. Bear in mind they may upon occasion have to be toggled from '1' to '0' for some installs or later when we get Also, this is a common practice when a developer needs to install and configure, **Node.js** behind a Corporate Web Proxy. **For further information, reference:** **a.** How to Setup Behind a Corporate Proxy: https://jjasonclark.com/how-to-setup-node-behind-web-proxy/ **b.** SSL Certificates Explained: https://medium.com/nodejs-tips/ssl-certificate-explained-fc86f8aa43d4
7	Common Practice to restart your computer here...	Restart Computer Now
8	Right-Click on the Desktop Shortcut **Node.js** and "**Run Elevated with DefendPoint**" ***It is a always good practice to run applications as "**Run Elevated with DefendPoint**"	Opens **Node.js Command Prompt**
9	To Continue the setup from **behind a "Corporate Web Proxy"** Update the "**Node.js Configuration File.**	Navigate to to where **Node.js** was installed. For me it was the following folder structure: "**C:\nodejs**" Type the following statements at the "**npm**" Command Line and hit the ENTER Key to update the "**Node.js**" Configuration File:. **a.** npm config set strict-ssl=false

		b. npm config set proxy=http://webproxy.igslb.allstate.com:8080/ **c.** npm config set https-proxy=http://webproxy.igslb.allstate.com:8080/
10	Make one additional update to the "**Node.js npm**" Configuration File.	Type the following statements at the "**npm**" Command Line and hit the ENTER Key. -- This will point **Node.js** at the **Allstate Artifactory Location** **a.** npm config set registry=https://artifactory.allstate.com/artifactory/api/npm/npm-cf – Use this Registry for AllState Configurations. For example, when installing APPFabric, this registry will need to be used. **b.** npm config set registry=https://artifactory.allstate.com:443/artifactory/api/npm/eis-allstate-npm/ - Use this Registry for Vendor/EIS Configurations. ***This can also be set by editing the "**.npmrc**" file found at the following folder structure; 
11	Install **Allstate APPFabric CLI & Package Dependencies**	Type the following statements at the "**Node.js**" Command Line and hit the ENTER Key. **a.** npm login This will log the developer into the **Artifactory** that was just configured as the **REGISTRY** in the previous **Step #10** Once successfully logged in, Type the following statements at the "Node.js" Command Line and hit the ENTER Key. **b.** npm install -g app-fabric-cli **T**his will install the **APPFabric** Framework to your local machine
12	Create System Environment Variables for **LOCAL_USER** & **LOCAL_PASSWORD**	Both System Environment Variables are the credentials necessary to access the **Artifactory**, mentioned in the previous step. Without them you will be unable to build projects. This will become important around **Step #37**, when we install the application build software "**Maven**" and start the process of building our application. ⚠ The values will be dependent upon the **Developers' Network Credentials** and **NOT** the "**System User ID & Password** " that was established for the project through **MarketPlace**. Reference the following for building an application: Team Confluence Page: Import an Existing Project & Build Team Confluence Page: Create a New Project & Build
13	Validate that the "**Node.js** " Configuration File has been updated.	Type the following statements at the "**npm**" Command Line and hit the ENTER Key. **a.** npm config edit We should be able to observe the npm configuration changes made within **Step #10** *****An Example Copy has been attached for the reader's reference*****

		text file
14	Add the two Certificates to npm	**a.** Open a Windows Folder Explorer and Navigate to your systems ".npmrc" File. This is located underneath your USERS NTID Folder as shown in the in the screen shot *below.(C:\Users**insertYourNTIDHere**\.npmrc)* This file contains the "npm" configuration contents that we observed within **Step #13.**  **b. O**pen the file for Editing **c.** Proceed to cut and paste the following certificates into the ".npmrc" File, at the top of the file. ca[]="-----BEGIN CERTIFICATE-----\nMIIFAzCCAuugAwIBAgIQZeY2Qz4gHI9OzkNUfOzZzzANBgkqhkiG9w0BAQsFADAU\nMRIwEAYDVQQDEwlBcmllc1Jvb3QwHhcNMTUwMjI1MTU1MjM: ca[]="-----BEGIN CERTIFICATE----- \nMIIHWjCCBUKgAwIBAgITEwAAAAIvvVhtXav/+wAAAAAAAjANBgkqhkiG9w0BAQsF\nADAUMRIwEAYDVQQDEwlBcmllc1Jvb3QwHhcNMTUwMzA3MDM1NjIzWhcNMzAwMzA3\nMDQwN -----END CERTIFICATE-----" **d.** Your file should not resemble something similar to the following; .npmrc
15	29 Jun 2021 - New Step Added ***Update your LOCAL Machines Host File with 'npm' Registry Settings**	This step is more of a precursor than anything. We have been experience Network Timeout's or ERRConnectionRefused Errors to the point where we have updated the LOCAL Machine Hos 151.101.36.162 registry.npmjs.com 151.101.36.162 registry.npmjs.org A copy of my Host File has been attached for reference; hosts

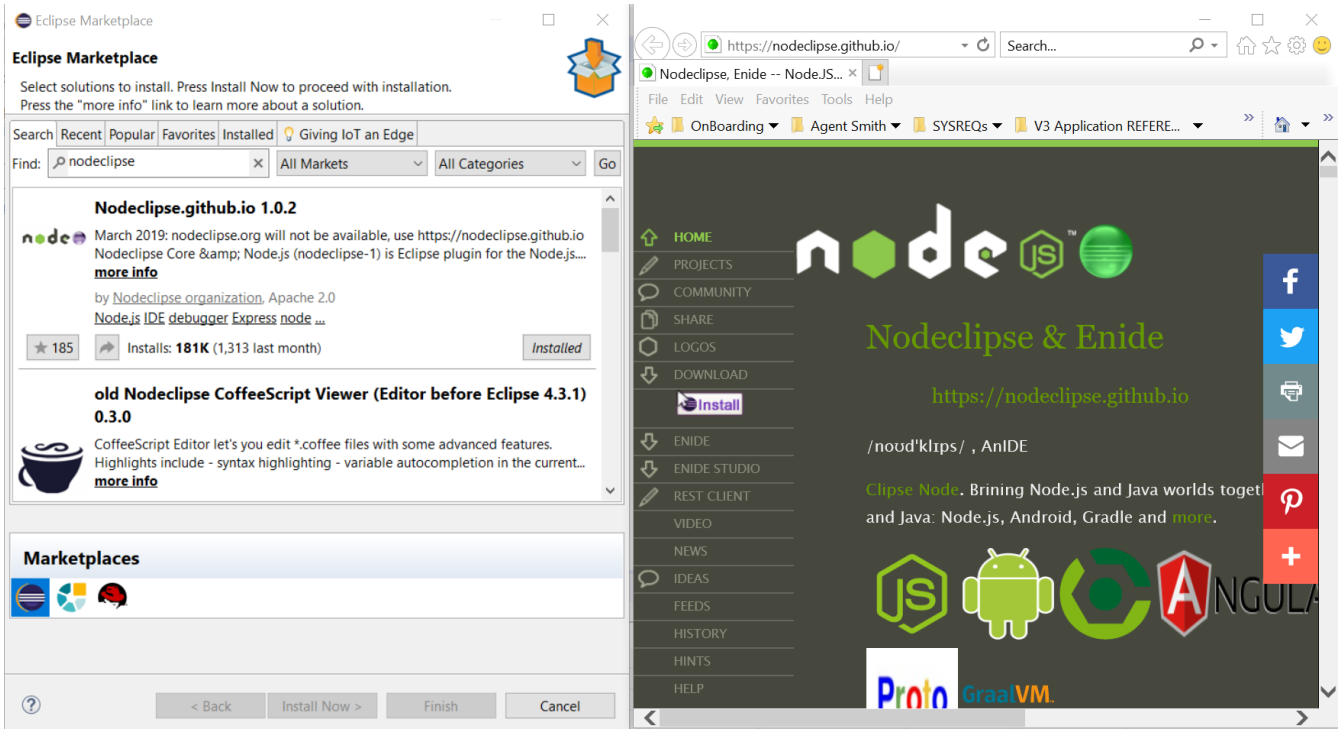
16	Install **Node.js Express Packages & Dependencies**	Type the following statements at the "**Node.js**" Command Line and hit the ENTER Key.
		a. npm install
		
		⚠ By default, npm install will install all modules listed as dependencies in package.json. Found beneath folder structure: ../node_modules/npm.
		This will automatically install all dependencies. If successful, there will be a series of Command Prompt Console Outputs written to the screen. Screen shot captured below for reference
		
		Select Administrator: c:\windows\system32\cmd.exe
		`glob@7.1.1 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\glob`
		`fs.realpath@1.0.0 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\fs.realpath`
		`inflight@1.0.6 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\inflight`
		`once@1.4.0 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\once`
		`wrappy@1.0.2 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\wrappy`
		`inherits@2.0.3 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\inherits`
		`minimatch@3.0.3 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\minimatch`
		`brace-expansion@1.1.11 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\brace-expansion`
		`balanced-match@1.0.0 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\balanced-match`
		`concat-map@0.0.1 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\concat-map`
		`path-is-absolute@1.0.1 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\path-is-absolute`
		`up to date in 2.739s`
		
		`17 packages are looking for funding`
		`run `npm fund` for details`
		
		`npm notice created a lockfile as package-lock.json. You should commit this file.`
		`added 399 packages from 349 contributors in 67.893s`
		
		`17 packages are looking for funding`
		`run `npm fund` for details`
		
		`C:\nodejs\node_modules\npm>`
		
		If the "npm" Command resulted in failure, install dependencies manually by executing following commands:
		• "npm install express -g"
		• "npm install url -g"
		• "npm install fresh -g"
		• "npm install cookie -g"
		• "npm install methods -g"
		• "npm install crc -g"
		• "npm install send -g"
		• "npm install connect -g"
		• "npm install commander -g"
		*****For additional Reference*****
		https://coderwall.com/p/mbov6w/running-nodejs-and-express-on-windows

17	Install **Nodeclipse CLI, Express, & Express Generator Package Dependencies**	Type the following global statements at the "**Node.js**" Command Line and hit the ENTER Key. **a.** npm install -g nodeclipse **b.** npm install -g express **c.** npm install -g express-generator ⚠ These are dependencies for the Eclipse IDE Nodeclipse Plugin mentioned later.

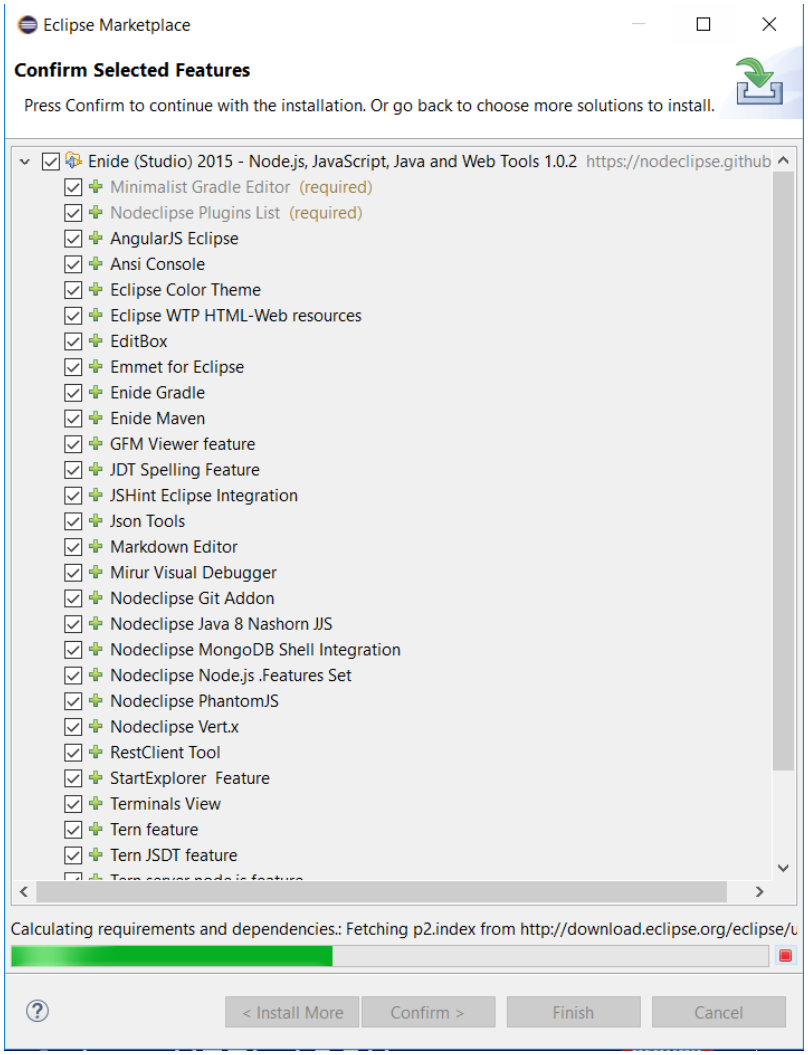
18	Download **JAVA**	Allstate Approved Version JAVA Version **11.0 +**, can be found at the Adopt OpenJDK's site; https://adoptopenjdk.net/ I chose to download the Windows 64 bit Version as an MSI File, ******It is critical that Step 1 was accomplished first in order to run this MSI File******  JAVA Platform Standard Edition 8 Documentation: https://docs.oracle.com/javase/8/docs/
19	Right-Click on the **OPENJDK11u-jdk_x64_windows_hotspot_11.0.11_9.msi** File and "**Run Elevated with DefendPoint**"	This will walk through a typical windows install process. The only nuance that I chose to update the install path to "**C:\Java**" from the default. "**C:\Program Files (x86)\Java\jre1.8.0+**". Run thi
20	Create/Update the **JAVA_HOME** & **PATH** System Environment Variables	Validate/Configure the **JAVA_HOME** & **PATH** System Environment Variable Paths have been set to the destination folder where the Java was installed to.
21	Download the latest version of Eclipse Integrated Development Environment (IDE)	⚠ As of this date, 31 May 2021 , the Eclipse IDE is within an Allstate PEAP Process, to receive permission to be used ON PREMISE. In the interim, have proceeded to download th As before, download the '**eclipse-inst-jre-win64**' 64 bit version; https://www.eclipse.org/
22	Right-Click on the **eclipse-inst-jre-win64.exe** Executable File and "**Run Elevated with DefendPoint**"	When the Eclipse Installer Window appears select the following: "**Eclispe IDE for JAVA Developers**" The following installation settings were selected: **a. Java 11.0.11.9+ VM:** "*C:\locationOfJAVA_HOME*" **b. Installation Folder: "C:\Eclipse**" from the default. "**C:\Users*userNTID*\eclipse\java-2020-03**". Run this to successful completion. **c.** Accept the License Agreement Once complete it will offer the following options: a. Launch b. Show README File c. Open in System Explorer d. Keep Installer I chose to Launch the IDE to continue with the configuration process.
23	Validate that the Eclipse IDE, JAVA Compiler and Installed JRE's are configured to **Step #'s 5 & 6**	Proceed to the following IDE Navigation: **Windows → Preferences → JAVA → Compiler** **Windows → Preferences → JAVA → Installed JRE's**
24	Install Nodeclipse Eclipse Plugin NOTE: Reference: the following link if necessary https://www.dev2qa.com/node-js-hello-world-example-with-eclipse/	**Navigate to the following:** **Help → Eclipse Market** Type "**Nodeclipse**" within the **Find Search Box**

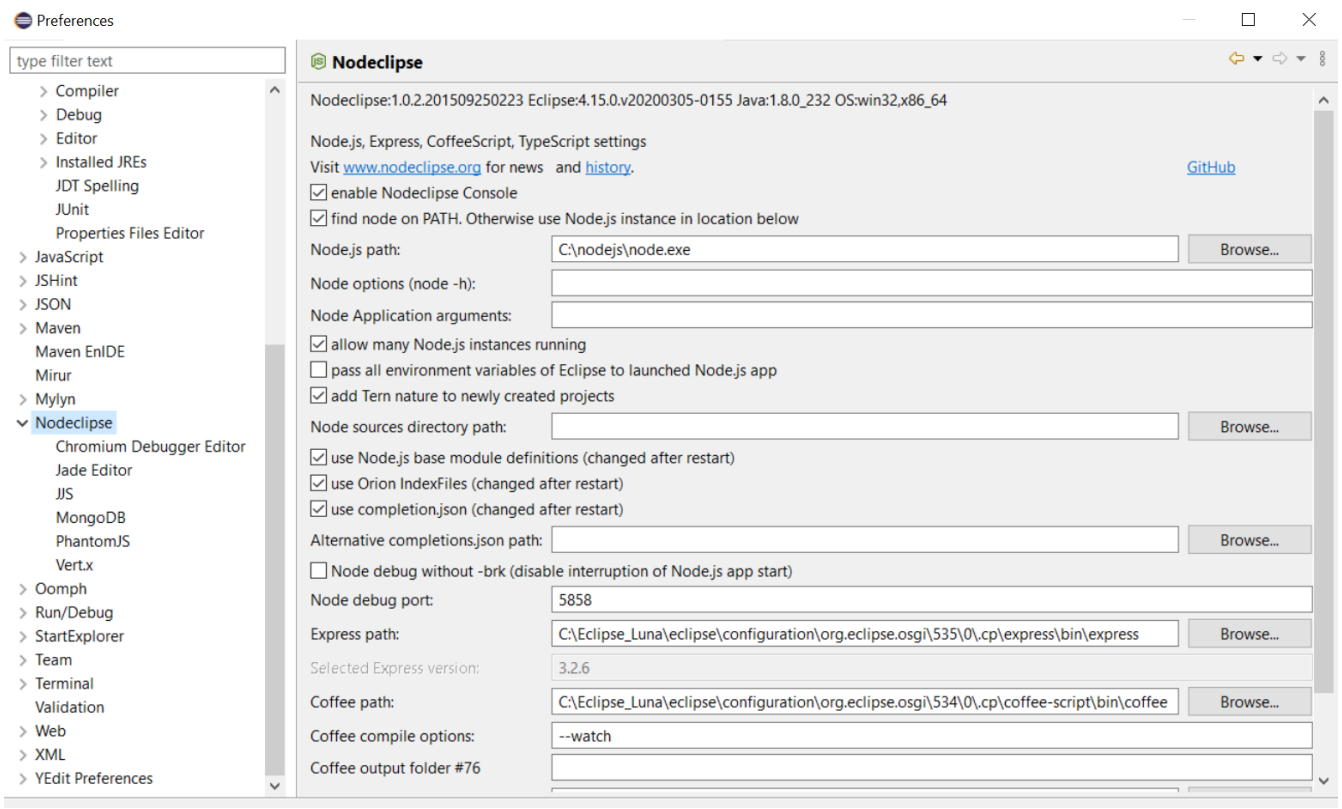
Hit the ENTER Key

The following will be returned, screenshot provided below. It will redirect the developer to a "**Nodeclipse & Enide**" Website; https://nodeclipse.github.io/



25	Drag and Drop the "**Nodeclipse & Enide**" Eclipse Plugin into MarketPlace;	Eclipse will start the process of plugin install. Select all features as shown below and continue.

		
26	Agree to the Install License	Agree to the Eclipse Plugin Install License and Continue
27	Restart Eclipse	Restart Eclipse.
28	Validate that Nodeclipse points at the newly installed version of "**Node.js**" Destination Folder (***Reference Step #3***)	Confirm Installed "**Node.js**" Destination Folder by performing the following navigation; **Window → Preferences → Nodeclipse** Reference Screen Shot Below.

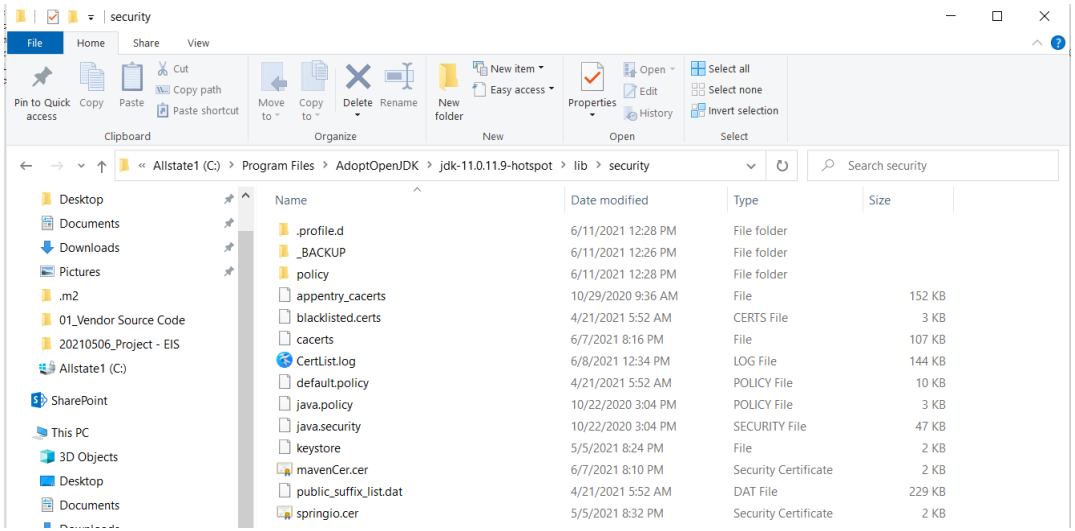


29	Install GitHub. **NOTE:** There are two options to install GIT. One is a Command Line Utility that I have personally chosen to download; whereas the other is a Desktop version of the application. If a previous version exists on your system, please uninstall before proceeding.	**GitHub Command Line Utility:** https://git-scm.com/book/en/v2/Getting-Started-Installing-Git **Installing on Windows** There are also a few ways to install Git on Windows. The most official build is available for download on the Git website. Just go to https://git-scm.com/download/win and the download will sta Download the Windows 64 bit Version as an EXE File. ******It is critical that Step 1 was accomplished first in order to run this EXE File****** ---------- **GitHub Desktop APP** can be found at the following link: https://desktop.github.com/ Download the Windows 64 bit Version as an EXE File. ******It is critical that Step 1 was accomplished first in order to run this EXE File******
30	Right-Click on the **Git-2.27.0-64-bit.exe** File and "**Run Elevated with DefendPoint**"	This will walk through a typical windows install process. If a previous version exists on your system, UNINSTALL it before proceeding. I have chosen to install to the destination folder, **C:\Git**, instead of the default. "**C:\Program Files\Git**". Run this to successful completion. Once successfully completed you may Click the "Next" Button It will redirect you to the Release Notes: file:///C:/Git/ReleaseNotes.html
31	Navigate to the following : Start → Programs → Git.	Right Click on **Git Bash and Open File Location**
32	Create Shortcut on Desktop for GitBash Command Prompt	File Explorer Window will Open. Right-Click on "Git Bash" and navigate to; **Send To → Desktop(Create Shortcut)**
33	Copy the KeyStore Certificate **cacerts** File out to your local JRE	Reference the following Keystore Certificate Project; https://github.allstate.com/Compozed/java-bp-cacerts

From this project, simply copy the **cacerts file** to your systems JAVA JRE system folder path. It has also been provided here for reference.

Make certain to backup the previous one prior to copying out the new certificate file.

> ⚠ Reference the "cacerts_Backup" as an example (Mine has been provided as an example and a screen shot follows);
>
> C:\Program Files\AdoptOpenJDK\jdk-11.0.11.9-hotspot\lib\security

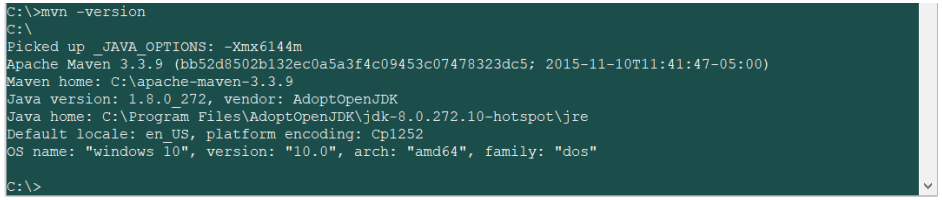
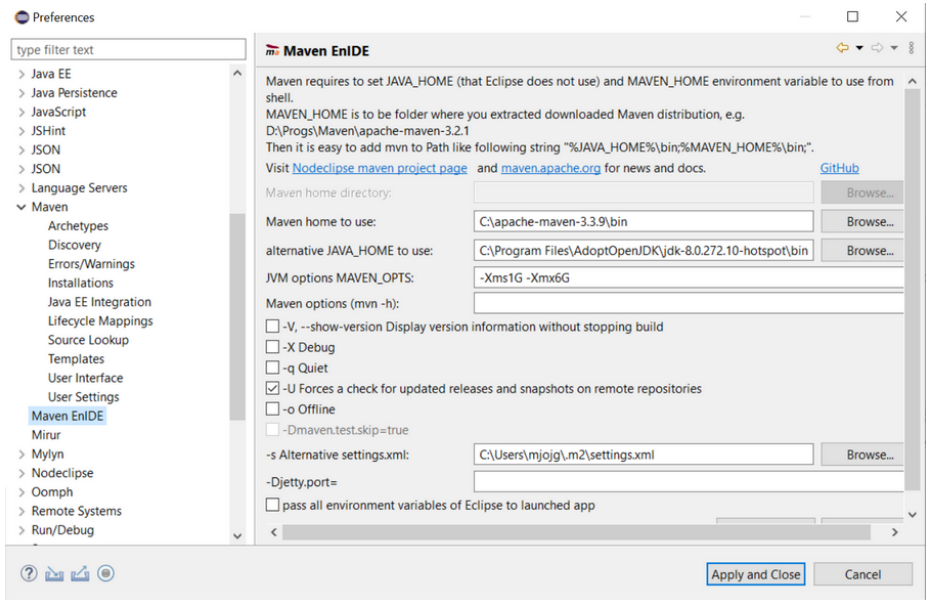


34	Prepare to configure **Git** andor **GitHub for a Corporate Web Proxy**	Based upon the project mentioned in **Step #28**, https://github.allstate.com/Compozed/java-bp-cacerts, the Keystore Certificate File, **cacerts** was then leveraged as a base to generate a Bund This Bundled Certificate File is what will be used to configure the Git and GitHub Web Proxy Settings. Both Bundled Files have been provided below. ⚠ ***For your reference what follows are the steps to create the Bundled Certificate File.*** ***Within the "certmgr" utility at the command line.*** It provides the means of **importing** the "cacerts" keystore file, only to then **export** it as a **Base64 Encoded Format** and **append it to the "ca-bundle.crt" File.** Also, from Google Chrome navigate to Allstate's GitHub Website, **https://github.allstate.com/** and **export** the **certificate** only to **repeat the process mentioned previously to appen** ***Within Google Chrome:*** ***a. Click on the menu (⋮) displayed on the top right corner after the browser address bar*** ***b. Now follow More tools >> Developer Tools.*** ***c. Select the Security tab, a second right option with default settings. Click on View Certificate and go to the "Details" you will have certificate details*** **For the Git Command Line Utility copy both bundles out to the following system folder path (*Remember this is specific to my machine and where I installed the Git Command Lir*** C:\Git\usr\ssl\certs **For the GitHub Desktop Application copy both bundles out to the following system folder path (*Remember this is specific to my machine, my user, mjojg, you will have to alter th*** C:\Users\mjojg\AppData\Local\GitHubDesktop\app-2.5.2\resources\app\git\usr\ssl\certs
35	Right-Click on the newly created Desktop Shortcut **GitBash** and "**Run Elevated with DefendPoint**" ***It is a always good practice to run applications as "**Run Elevated with DefendPoint**"	Should open a GitBash Terminal

36	Configure Git for your profile	Let's make sure that you have git installed and perform the initial configuration. Don't forget to substitute my name, email, and project with your values. Reference, https://chronicler.tech/windows-password-manager-and-git/, for further information. Execute the following GitBash Statements. Hit the ENTER Key after each statement **a.** git --version **b.** git version 2.27.0.windows.1 **c.** git config --global user.email ***insertEmailAddressHere*** **d.** git config --global user.name "***insertFirstLastNameHere***" **e.** git config --system core.longpaths true **f.** git config --system core.autocrlf false

37	Configure Git for the Corporate Web Proxy	Execute the following GitBash Statements. Hit the ENTER Key after each statement.
		
		If using the Git Command Line Utility Perform **Steps a & b,** this will allow you to enter your credentials upon using Git. For further information please reference the following; https://chronicl
		Please note this will be specific to your user, so please replace '***DOMAIN\\mjojg***' with your username.
		a. git config --global http.proxy http://DOMAIN\\mjojg@github.allstate.com
		b. git config --global https.proxy http://DOMAIN\\mjojg@github.allstate.com
		
		If using the GitHub Desktop Application Perform **Steps c & d,** this will clear the proxy settings
		c. git config --global http.proxy ""
		d. git config --global https.proxy ""
		
		Perform the remaining **Steps e & f**
		e. git config --global http.sslverify false
		f. git config --global http.sslCAInfo ***C:/Git/usr/ssl/certs/ca-bundle.crt***
		
		I chose to use the GitHub Desktop Application, so when finished your ***.gitconfig file***, located at your User System Folder path, mine has been provide for reference; ***C:\Users\mjojg,*** should r
		`[filter "lfs"]`
		`  process = git-lfs filter-process`
		`  required = true`
		`  clean = git-lfs clean -- %f`
		`  smudge = git-lfs smudge -- %f`
		`[user]`
		`  name = Maurice Johnson`
		`  email = mjojg@allstate.com`
		`[http]`
		`  proxy =`
		`  sslverify = false`
		`  sslCAInfo = C:/Git/usr/ssl/certs/ca-bundle.crt`
		`[https]`
		`  proxy =`
		`  sslverify = false`
		`  sslCAInfo = C:/Git/usr/ssl/certs/ca-bundle.crt`

38	Download the latest version of Maven and Install	As before, download the '**apache-maven-3.5.4-bin.zip**' 64 bit version or latest; https://maven.apache.org/ Unzip the archive to a system folder similar to the following: ***C:\apache-maven-3.5.4***
39	Create/Update the **MAVEN_HOME** & **PATH** System Environment Variables	Update the System Environment Variable **MAVEN_HOME** & **PATH** with the following ""
40	Validate successful installation	Right-Click on the Command Prompt Desktop Shortcut and "**Run Elevated with DefendPoint**" Execute the following statement: **a.** mvn -version

		 C:\>mvn -version C:\ Picked up _JAVA_OPTIONS: -Xmx6144m Apache Maven 3.3.9 (bb52d8502b132ec0a5a3f4c09453c07478323dc5; 2015-11-10T11:41:47-05:00) Maven home: C:\apache-maven-3.3.9 Java version: 1.8.0_272, vendor: AdoptOpenJDK Java home: C:\Program Files\AdoptOpenJDK\jdk-8.0.272.10-hotspot\jre Default locale: en_US, platform encoding: Cp1252 OS name: "windows 10", version: "10.0", arch: "amd64", family: "dos" C:\>
41	Close out all applications and restart your computer	Close out all applications and restart your computer
42	Double Click the Eclipse IDE Desktop Icon	This should open up the Eclipse IDE
43	Proceed to Eclipse Market Place and Validate that the following Plugins have been installed. If not please proceed to install a current version of each plugin.	Eclipse Plugins: *- ANSI Escape in Console* *- Eclipse Tools for Cloud Foundry* *- Eclipse Web Developer Tools* *- Eclipse Tomcat Plugin* *- Eclipse JST Server Adapters* *- Eclipse Enterprise Java and Web Developers* *- Eclipse Docker Tooling* *- Wild Web Developer (WWD)* *- Nodeclipse/Enide Maven for Eclipse* *- Nodeeclipse.github.io* *- Buildship Gradle Integration* *- Minimalist Gradle Editor* *- Eclipse Zip Editor*
44	Configure Project Maven with the following Settings; **Windows - Preferences - MAVEN**	⚠ Reference the following Research; EIS Version 12 - Eclipse IDE with Maven Builds - Runtimes Analyzed Dependent upon your LOCAL Machine, some configurations may not be suitat  **<u>Text of Snapshot:</u>** **Maven Home:** C:*locationOfWhereYourLOCALMavenIsInstalled* ***JAVA Home:*** *C:\locationOfWhereYourLOCALJAVAIsInstalled* **JVM Options:** -Xms1G -Xmx9G

		Vendor Recommended JVM Options: -Xms4G -Xmx12G -XX:+UseParallelGC -XX:MetaspaceSize=2000M -XX:MaxMetaspaceSize=3000M -Duser.country=US -Duser.region=US -Duser.la **Alternative Settings.xml***: C:\Users\yourNTID\.m2\settings.xml* **Check** *'-U Forces a check for updated releases and snapshots on remote repositories'* **Uncheck** *'Download Artifact Sources' (Found at the highest level,* ***Maven****)*
45	**Congrats!!! You are now ready to Import an Existing Maven Project and Build it..**	

No labels